CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 5: Dijkstra Single-Source Shortest Path Algorithm

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #5 | Academic Year: 2026-27

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1. Overview & Objective:

Compute shortest path routing trees on weighted directed graphs with non-negative edge weights.

2. Practical Tasks & Verification Steps:

1. Initialize distance array with infinity and source vertex distance to 0.
2. Maintain Min-Heap PriorityQueue extracting vertex with smallest tentative distance.
3. Relax all outgoing edges from extracted vertex updating distance array when shorter path is discovered.
4. Reconstruct and print the complete shortest path sequence from source to all target vertices.

Submission Guidelines & Verification:

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